DECAF-CD: A Semisupervised Dual-Encoder Channel-Aware Network for Fine-Grained Forest Change Detection

Jinghe Zhou, Jinmei Shi, *Member, IEEE*, XianPeng Wang, *Member, IEEE*, Xingwang Li, *Senior Member, IEEE*, Jianglin Luo, and Yu-Beng Leau, *Senior Member, IEEE*

***Abstract*—Accurate forest change detection (CD) is crucial for ecosystem health monitoring, deforestation and degradation assessment, and sustainable large-scale forest management. However, forest change detection remains highly challenging due to strong spectral similarity among vegetation types, irregular and fragmented geometry of disturbances, blurred canopy boundaries, and a severe scarcity of high-quality annotations. Acquiring labeled data is costly and labor-intensive, typically requiring expert interpretation of high-resolution imagery, which limits the scalability and generalization of fully supervised models. To address these limitations, we propose dual-encoder channel-aware forest change detection network (DECAF-CD), a semisupervised deep learning framework for fine-grained forest CD. This framework jointly leverages temporal alignment, multiscale structural aggregation, and boundary-aware modeling to capture subtle and irregular forest disturbances while effectively utilizing unlabeled data. Specifically, the framework first, aligns bitemporal features to suppress pseudochanges, second, aggregates multiscale structural cues to distinguish complex disturbance patterns, and third, refines object boundaries to enhance spatial coherence, all within a semisupervised learning paradigm that significantly reduces reliance on scarce annotations. Extensive experiments on five benchmark datasets show that DECAF-CD outperforms seven state-of-the-art semisupervised CD methods consistently across annotation ratios from 5% to 40%. The model even surpasses fully supervised methods, while further validating the strong generalization capability of the proposed framework. Notably, DECAF-CD lowers annotation requirement by as much as 87.5% for all evaluated datasets. The largest performance improvements are seen on the RF-CD dataset, where weak spectral contrast and irregular boundary structures make the detection of fine-grained forest disturbances especially difficult, which highlights the effectiveness of our method in tackling the challenging forest scenarios.**

***Index Terms*—Change detection (CD), channel-aware, dual-encoder, semisupervised.**

I. Introduction

ACCURATE monitoring of forest disturbances is essential for assessing ecosystem health, evaluating the impacts of deforestation and forest degradation, and supporting sustainable forest management. The growing accessibility of high-resolution remote sensing imagery, coupled with advances in deep learning, presents unprecedented opportunities for automated forest [1]. Unlike traditional manual delineation and on-site surveys, which are characterized by labor intensity, time consumption, and difficulty in scaling, automated approaches are capable of facilitating continuous, large-scale monitoring, and timely detection of both natural disturbances and anthropogenic activities [2]. These technological advances lay a solid foundation for addressing the intricate spatiotemporal dynamics inherent in forest ecosystems, which remain challenging for conventional remote sensing techniques.

Forest change detection (CD) presents unique challenges that are distinct from urban or built-up change analysis. First, forest landscapes are highly heterogeneous and exhibit complex spatial patterns, with disturbances spanning a wide spectrum, from selective logging and narrow access roads to canopy gaps and large clear-cuts [3]. Second, spectral variability caused by seasonal phenology and illumination effects complicates the discrimination between genuine changes and pseudochanges, particularly among forests, low vegetation, and croplands [4]. Third, changes occur across multiple spatial scales, requiring models capable of capturing both fine details and broader contextual structures [5]. Finally, and most critically, annotated datasets for forest disturbances are extremely limited, especially for rare

Received 7 December 2025; revised 22 February 2026; accepted 21 March 2026. Date of publication 24 March 2026; date of current version 21 April 2026. This work was supported in part by the "New Star of the South China Sea" Science and Technology Innovation Talent Platform Project in Hainan Province of China under Grant NHXXRCXM202321 and Grant JYNHXX2023-14Q and in part by the Natural Science Foundation of Hainan Province in China under Grant 624QN283. *(Corresponding authors: Jinmei Shi; XianPeng Wang.)*

Jinghe Zhou is with the Faculty of Computing and Informatics, Universiti Malaysia Sabah, Kota Kinabalu 88400, Malaysia, also with the School of Information Engineering, Hainan Vocational University of Science and Technology, Haikou 570100, China, and also with the Hainan Province Wisdom Internet of Things Engineering and Technology Research Center, Hainan Vocational University of Science and Technology, Haikou 570100, China (e-mail: chaujhe@163.com).

Jinmei Shi and Jianglin Luo are with the School of Information Engineering, Hainan Vocational University of Science and Technology, Haikou 570100, China, and also with the Hainan Province Wisdom Internet of Things Engineering and Technology Research Center, Hainan Vocational University of Science and Technology, Haikou 570100, China (e-mail: Shijinmei1003@hvust.edu.cn; leal_ljl@163.com).

XianPeng Wang is with the School of Information and Communication Engineering, Hainan University, Haikou 570228, China (e-mail: wxpeng2016@hainanu.edu.cn).

Xingwang Li is with the School of Physics and Electronic Information Engineering, Henan Polytechnic University, Jiaozuo 454000, China (e-mail: lixingwangbupt@gmail.com).

Yu-Beng Leau is with the Faculty of Computing and Informatics, Universiti Malaysia Sabah, Kota Kinabalu 88400, Malaysia (e-mail: lybeng@ums.edu.my).

Digital Object Identifier 10.1109/JSTARS.2026.3677133

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or small-scale events, which severely restricts the effectiveness and generalization capacity of purely supervised approaches [6]. Among these challenges, the scarcity of labeled data represents the primary bottleneck for advancing forest CD.

Existing deep learning-based CD methodologies have demonstrated exceptional performance in urban and land-cover monitoring endeavors [7]. However, these methods are frequently devised for general CD or fully supervised settings and fail to address the specific complexities of forest environments [8]. In particular, few frameworks explicitly confront the combined challenges posed by irregular forest morphologies, multiscale heterogeneity, ambiguous boundaries, and scarce annotations. Consequently, a substantial research gap persists for the development of a dedicated, annotation-efficient, semisupervised framework that tailored for fine-grained forest CD.

To address this gap, we introduce a semisupervised deep learning framework specifically designed for forest CD. The proposed network amalgamates three pivotal modules to surmount the aforementioned challenges: a dual-branch encoder equipped with global attention fusion (GAF) mechanism, which is designed for temporal alignment and suppression of pseudochanges; a cross-scale group-concat (CGC) module, augmented with large kernel attention (LKA), to capture irregular and multiscale forest disturbances while enforcing boundary consistency; and a multipath dilated decoder (MDD) to progressively refine multiscale features, thereby enabling precise delineation. Furthermore, the framework is further fortified by incorporating a teacher–student semisupervised learning paradigm, which empowers the network to leverage unlabeled data and attain superior performance with minimal annotated samples.

The principal contributions of this work are summarized as follows.

1) We propose dual-encoder channel-aware forest change detection network (DECAF-CD), an annotation-efficient semisupervised framework tailored for forest CD, which attains performance levels comparable to fully supervised approaches, with only a small fraction of labeled data, thereby substantially reducing the annotation workload.
2) We design a GAF module specifically engineered for channel-level temporal alignment, which effectively mitigates pseudochanges induced by illumination, shadows, and phenological variations.
3) We develop a CGC module integrated with LKA to capture irregular and multiscale disturbance patterns while enhancing boundary consistency.
4) Extensive experiments on benchmark datasets substantiate the efficacy of the proposed framework, achieving competitive performance with a reduction of up to 87.5% in the number of labeled samples required, thereby highlighting its practical viability for large-scale forest monitoring.

II. Related Work

A. Deep Learning for Remote Sensing CD

The emergence of deep learning has revolutionized the paradigm of remote sensing CD [9]. Convolutional neural networks (CNNs) and encoder–decoder architectures have been extensively employed in bitemporal imagery, enabling hierarchical feature extraction and robust modeling of spatial patterns [10]. Compared with traditional pixel-wise or object-based methods, deep learning models possess the capability to discern intricate semantic discrepancies between pre- and postevent images, improving the detection of subtle or heterogeneous changes. The integration of attention mechanisms, multiscale feature aggregation, and feature fusion strategies has further enhanced the discriminative capacity of these networks, allowing them to capture long-range dependencies and subtle structural variations. Nevertheless, the majority of early-stage methods were designed for urban or anthropogenic environments, where targets exhibit homogeneous textures, regular geometries, and spatially discrete changes. When deployed in forest ecosystems, these models frequently encounter challenges in effectively managing irregular morphologies, multiscale disturbances, and diffuse boundaries, thereby limiting their practical applicability for large-scale ecological monitoring.

B. Challenges in Forest CD

Forest CD presents a series of distinctive challenges that are largely absent in urban or built-up areas. First, spectral variability stemming from seasonal phenology or environmental conditions can induce pseudochanges, consequently resulting in elevated false-positive rates [11], [12]. Second, natural disturbances, such as selective logging, wildfires, pest infestations, and canopy regeneration, generate irregular, fragmented, and elongated spatial patterns, thereby complicating precise boundary delineation [6]. Third, forest disturbances manifest across multiple scales, ranging from narrow logging roads to extensive clear-cuts, requiring models to concurrently capture both fine- and coarse-grained structures features [13], [14]. Finally, the scarcity of accurately annotated datasets, particularly for rare or small-scale disturbances, constrain the effectiveness of fully supervised methods and challenges model generalization [15], [16]. These factors emphasize the need to develop architectures that are both structurally cognizant and temporally consistent while remaining efficient under semisupervised conditions.

C. Multiscale and Boundary-Aware Methods

To address the challenges posed by heterogeneous and multiscale change patterns, recent approaches have incorporated multiscale feature fusion, attention-based modeling, and boundary enhancement. Multiscale networks amalgamate low-level spatial details with high-level semantic context, enabling detection of both minor and extensive changes [17], [18]. Edge-aware modules or mechanisms with large receptive fields improve boundary precision, enabling the capture of irregular and diffuse forest edges [19], [20]. Although these methods advance the state-of-the-art, the majority of existing architectures either concentrate on a singular aspect, such as multiscale aggregation and edge modeling, or rely heavily on labeled data and postprocessing procedures, which consequently restricts their robustness and annotation efficiency in complex natural environments.

Channel attention mechanisms, such as SENet, recalibrate channel-wise feature responses for single-image

representations but fail to account for the cross-temporal nature inherent in bitemporal CD. The proposed GAF module extends this paradigm to concatenated bitemporal feature maps, enabling temporal-aware channel recalibration that adaptively suppresses pseudochanges arising from illumination shifts and seasonal variations. For cross-scale feature fusion, existing methods, such as FPN and ASPP aggregate multiscale context independently at each scale, without explicitly modeling the elongated and anisotropic spatial patterns characteristic of forest disturbances. The LKA offers an efficient mechanism for capturing long-range dependencies through decomposed depth-wise operations. However, it has not been applied within a grouped cross-scale fusion context. The proposed CGC module addresses this gap by embedding LKA into a grouped cross-scale concatenation framework, specifically designed to capture the long-range spatial dependencies along irregular forest boundaries that isotropic single-scale convolutions are inherently limited in representing.

D. Semisupervised Learning for Limited Annotations

The scarcity of labeled datasets has motivated the adoption of semisupervised learning strategies within the realm of remote sensing CD. Teacher–student paradigms, consistency regularization, and pseudolabeling techniques have been employed to leverage unlabeled samples, thereby effectively diminishing annotation requirements while maintaining model performance [21], [22]. These approaches have demonstrated promising results in urban and building-change datasets. However, their efficacy in highly heterogeneous and spectrally ambiguous forest ecosystems remains constrained. The absence of explicit structural priors in numerous semisupervised frameworks can lead to unstable predictions or loss of fine-grained details when addressing irregular natural disturbances [23], [24].

E. Motivation for the Proposed Framework

Building upon these observations, an effective forest CD model must jointly address temporal misalignment, spectral ambiguity, multiscale heterogeneity, and boundary irregularity, while efficiently leveraging limited labeled data. Our framework integrates these requirements into a unified architecture by combining dual-branch feature extraction, GAF for temporal alignment, CGC enhanced with LKA for structural and boundary-aware feature aggregation, and an MDD for progressive refinement of change predictions (CPs). The semisupervised teacher–student strategy further ensures robust learning in low-annotation scenarios. This integrated design explicitly targets the key challenges identified in Sections II-B–II-D, establishing a clear foundation for the detailed architectural description in Section III.

III. Material and Methods

A. Overall Architecture

The proposed DECAF-CD is a semisupervised framework designed for fine-grained forest CD under limited labeled data and highly heterogeneous forest environments. As illustrated in Fig. 1, the framework processes pre- and postevent images that are coregistered using a dual-encoder backbone to extract multilevel temporal features. Channel blocks are used to refine the feature representations by increasing discriminative responses and reducing the irrelevant information. Afterward, a GAF module restructures and fuses cross-level features. The fused features are further modeled by a CGC module coupled with LKA, which allows good multiscale contextual aggregation and spatial dependency modeling. The resulting feature representations are then fed to an MDD module that focuses on temporal discrepancies between bitemporal inputs. Finally, a segmentation head is used to generate the fine-grained forest disturbance maps, which are followed by postprocessing refinement to enhance the spatial consistency and boundary delineation. To further improve robustness and reduce reliance on dense annotations, DECAF-CD adopts a teacher–student semisupervised strategy, which enables the model to leverage abundant unlabeled bitemporal images through pseudolabel guidance and consistency regularization.

Given a pre-event image I_{pre} and a postevent image I_{post}, the dual encoders extract multilevel feature representations

$$\left\{F_l^{\text{pre}}, F_l^{\text{post}}\right\}_{l=1}^{4} = \mathcal{E}(I_{\text{pre}}, I_{\text{post}}). \tag{1}$$

These features are processed by the GAF module to produce adaptive discrepancy-aware representations that emphasize temporal changes while suppressing unchanged background. The refined features are subsequently enhanced by the CGC-LKA module, where channel reweighting and large-kernel fields jointly enhance disturbed forest structures and boundary clarity. Ultimately, the MDD modulea multidimensional decoder progressively aggregates cross-scale information to reconstructs a fine-grained change map $\hat{Y}$.

During the semisupervised learning, the teacher network $\mathcal{T}$ generates soft pseudolabels for unlabeled samples

$$\tilde{Y} = \mathcal{T}\left(\{\mathcal{F}_l^{\text{pre}}, \mathcal{F}_l^{\text{post}}\}_{l=1}^{4}\right), \quad \tilde{Y} \in [0,1]^{H\times W} \tag{2}$$

while the student network $\mathcal{S}$ is optimized through the utilization of both supervised and unsupervised consistency losses. The teacher parameters are updated via exponential moving average (EMA)

$$\theta_T \leftarrow \alpha\theta_T + (1-\alpha)\theta_S. \tag{3}$$

This EMA-based update stabilizes pseudolabel quality and circumvents confirmation bias. This integrated framework enables DECAF-CD to effectively learn disturbance patterns, sharpen boundaries, and generalize across diverse forest ecosystems despite limited manual annotations.

B. Dual-Encoder Feature Extraction and GAF

Discerning authentic forest disturbances from seasonal or phenological variations is arduous due to the interference of illumination, shadow, and atmospheric noise. The acquisition of both fine-grained spectral details and high-level semantic context is imperative for mitigating pseudochanges and retaining pertinent temporal information [25].

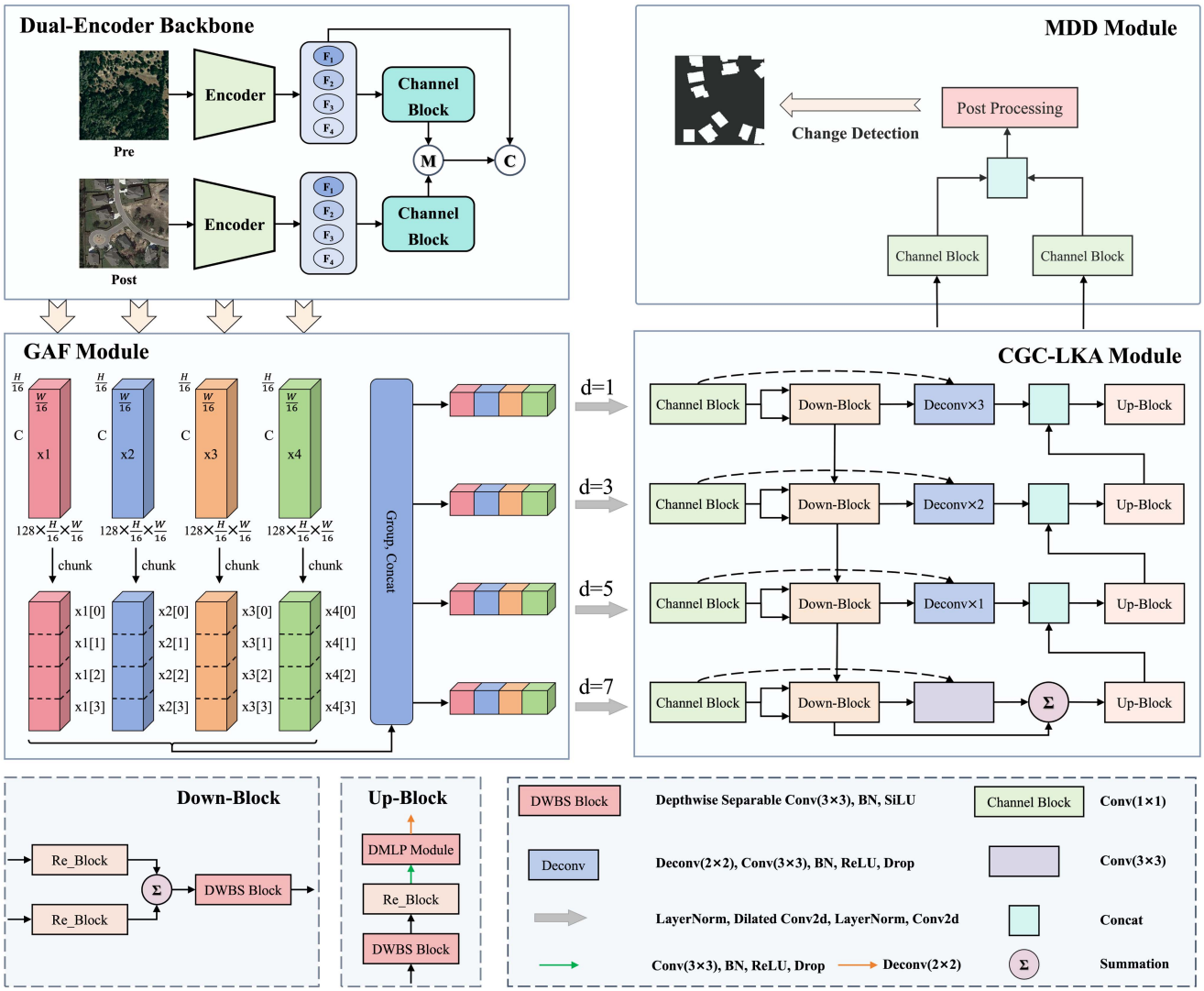


Fig. 1. Overview of the proposed DECAF-CD network. The framework integrates a dual-encoder backbone for bitemporal representation learning, a GAF module for cross-temporal interaction, a CGC block with LKA for global-local semantic enhancement, and a multidimensional decoder called MDD for high-resolution reconstruction.

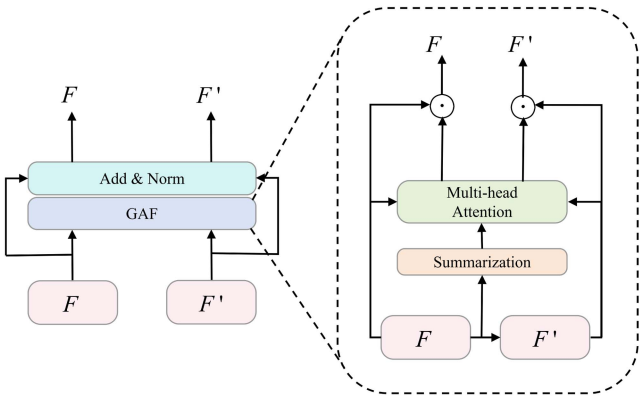


Fig. 2. Architecture of the GAF module.

As presented in Fig. 2, DECAF-CD employs a dual-branch encoder with shared weights to extract multilevel bitemporal features

$$F_i, F_i' = E(I_{\text{pre}}), E(I_{\text{post}}), \quad i = 1, \ldots, 4 \tag{4}$$

where i indexes the feature scale, F_i and F_i' denote the pre- and postevent feature maps with channel dimension C_i and spatial dimensions $H_i \times W_i$. To enhance temporal consistency and suppress pseudochanges, the GAF module concatenates features channel-wise

$$\tilde{Z}_i = [F_i, F_i'] \in \mathbb{R}^{2C_i \times H_i \times W_i} \tag{5}$$

and generates a channel-wise attention map via global pooling and softmax normalization

$$\alpha_i = \text{Softmax}\big(\text{Pool}(\tilde{Z}_i)\big) \in \mathbb{R}^{2C_i \times 1 \times 1}, \quad \hat{Z}_i = \tilde{Z}_i \odot \alpha_i \tag{6}$$

where $\odot$ denotes element-wise multiplication. Finally, the fused features are projected through a channel block

$$x_i = \sigma(W_i \hat{Z}_i + b_i) \tag{7}$$

where W_i, b_i are learnable weights and σ is the SiLU activation function. All fused features $\{x_1, x_2, x_3, x_4\}$ are resized to a unified spatial resolution $H/16 \times W/16$ for subsequent multiscale processing.

The dual-encoder with GAF provides discrepancy-aware and temporally aligned features that effectively distinguish genuine forest disturbances while suppressing pseudochanges from seasonal or atmospheric variations.

C. CGC-LKA Module

Forest disturbances exhibit irregular and multiscale heterogeneity. Capturing long-range dependencies and aligning multiscale features are crucial for detecting subtle changes and enhancing boundary delineation [26], [27].

The structure of the LKA module is shown in Fig. 3. Each fused feature x_i is refined via LKA, which decomposes a large convolutional kernel into depth-wise and point-wise operations to capture long-range spatial dependencies

$$x_i \leftarrow \mathrm{LKA}(x_i). \tag{8}$$

The refined features are then partitioned into four nonoverlapping channel groups

$$x_i = \{x_i[0], x_i[1], x_i[2], x_i[3]\},$$
$$x_i[j] \in \mathbb{R}^{C/4 \times H/16 \times W/16} \tag{9}$$

and cross-scale concatenated within each group j

$$G_d = \mathrm{Concat}[x_1[j], x_2[j], x_3[j], x_4[j]]$$
$$j = d - 1,\ d \in \{1, 3, 5\}. \tag{10}$$

The CGC-LKA module integrates shallow, intermediate, and deep-level features to enhance sensitivity to irregular disturbance shapes, sharpen boundary delineation, and align multiscale contextual information.

D. MDD Module

Forest CD inherently involves a tradeoff between boundary precision and semantic completeness. High-resolution features preserve fine-grained disturbance edges. Low-resolution features encode broader contextual semantics, which are essential for disambiguating spectrally similar land-cover transitions [28], [29]. The proposed MDD addresses this challenge by constructing four parallel decoding paths with dilation rates $d \in \{1, 3, 5, 7\}$. The spacing with odd intervals among these rates minimizes feature redundancy between adjacent paths, while enabling a systematic expansion of the effective receptive field across paths.

The structure of the MDD module is illustrated in Fig. 4. Each scale-hybrid group G_d from the CGC module is first processed through a channel block and a down-block to generate skip features S_d, which capture local structural context at the corresponding scale. Subsequently, the four decoding paths are executed in a serial top-down cascade. For paths $d \in \{1, 3, 5\}$, each up-block integrated with a DMLP (detailed in Section III-E) fuses the deconvolved feature map, the corresponding skip

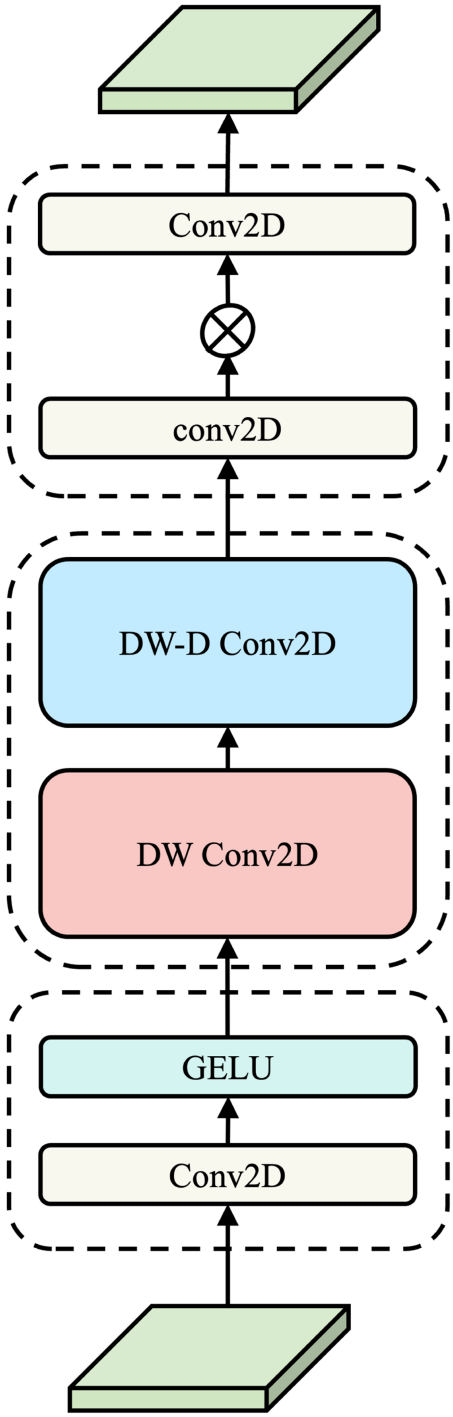


Fig. 3. Architecture of the LKA module.

features S_d, and the output of the preceding path H_{d-1}

$$H_d = U_d(\mathrm{Deconv}(S_d), S_d, H_{d-1}) \tag{11}$$

where H_{d-1} is initialized to zero for $d = 1$. Through this serial cascade, each H_d implicitly encodes the accumulated contextual information from all preceding paths. For the final path $d = 7$, H_5 already carries the aggregated multiscale representation of all prior paths. Therefore, broader context is further incorporated via element-wise summation rather than concatenation to avoid

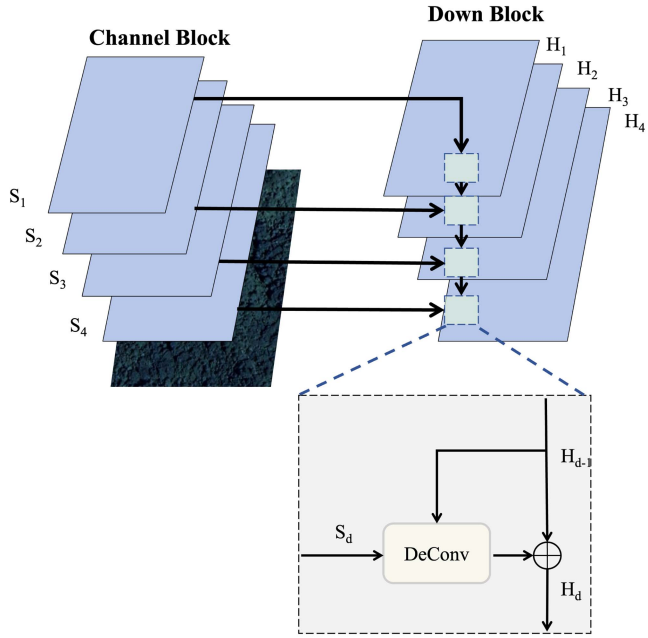


Fig. 4. Architecture of the MDD module.

inflating channel dimensionality

$$H_7 = U_7(S_7 + H_5, S_7). \tag{12}$$

The outputs H_1, H_3, H_5, and H_7 are subsequently concatenated within the MDD to form the final multiscale representation $H_{\text{final}} = \text{Concat}[H_1, H_3, H_5, H_7]$, which is passed to the segmentation head for change map generation. This cascaded multipath design enables MDD to progressively integrate fine boundary details from shallow paths with broad semantic context from deeper paths, providing a principled mechanism to delineate the irregular and diffuse disturbance boundaries characteristic of forest CD.

E. Core Computational Blocks

Efficient and discriminative feature extraction is essential for high-resolution satellite imagery. Specialized blocks include depthwise separable convolution with BN and SiLU for parameter efficiency (DWBS), down-block (two Re Blocks + DWBS for robust local features), and up-block with DMLP (fusing skip connections with upsampled features using dilated convolutions):

$$y = \text{Conv2d}(\text{LayerNorm}(\text{DilatedConv}(x, d))) \tag{13}$$

where the dilation matches the corresponding path to adaptively capture both fine-scale and broad contextual information. These blocks balance computational efficiency with expressive power, enabling the decoder to capture multiscale textures and semantics.

F. Semisupervised Teacher–Student Strategy

Limited labeled data motivates the utilization of unlabeled bitemporal images to improve generalization and reduce overfitting [30], [31], [32].

The teacher network $\mathcal{T}$ generates soft pseudolabels for unlabeled samples

$$\hat{Y}_t = \mathcal{T}(I_{\text{pre}}, I_{\text{post}}) \tag{14}$$

and the student network $\mathcal{S}$ is trained with a combination of supervised and unsupervised consistency losses

$$\mathcal{L}_{\text{semi}} = \mathcal{L}_{\text{sup}}(Y, \hat{Y}_s) + \lambda \mathcal{L}_{\text{cons}}(\hat{Y}_t, \hat{Y}_s). \tag{15}$$

Teacher parameters are updated using EMA

$$\theta_T \leftarrow \alpha\theta_T + (1-\alpha)\theta_S. \tag{16}$$

This semisupervised strategy enables DECAF-CD to exploit unlabeled bitemporal imagery while maintaining stable pseudolabel quality.

G. Change Map Generation and Loss Function

Accurate CD requires boundary-aware supervision and robust semisupervised learning [33].

The decoder outputs from all paths H_1, H_3, H_5, H_7 are concatenated

$$H_{\text{final}} = \text{Concatenate}[H_1, H_3, H_5, H_7]. \tag{17}$$

Two 1×1 convolutions then reduce the channel dimension to produce a single-channel logit map Y. A postprocessing step removes small spurious regions. The overall loss integrates binary cross-entropy (BCE), dice loss, and teacher–student consistency loss

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(Y, \hat{Y}_s) + \beta \mathcal{L}_{\text{Dice}}(Y, \hat{Y}_s) + \lambda \mathcal{L}_{\text{consistency}}(\hat{Y}_t, \hat{Y}_s). \tag{18}$$

This objective ensures the generation of accurate, boundary-preserving change maps, and facilitates effective semisupervised learning for data-scarce forest-CD.

IV. Experiments Result and Analysis

This section gives a full evaluation of the experimental evaluation of the proposed method. We start with introducing the benchmark datasets, semisupervised experimental protocol, and detailed implementation settings, followed by a description of the evaluation criteria and metrics. Subsequently, we show extensive comparisons with state-of-the-art semisupervised CD approaches. In addition, detailed ablation studies are performed in order to evaluate the contribution of each of the components, as well as a comparative study of the computational complexity.

A. Datasets

Experiments were performed on three forest-specific remote sensing CD datasets to fully assess the performance of the proposed algorithm in forest environments. In addition, two urban building CD datasets were used to conduct ablation studies and further validate the effectiveness of individual components.

1) *RF-CD* is a multitemporal satellite image dataset derived from publicly available Google Earth imagery, whose public availability has greatly enabled a wide range of remote sensing research [34]. The dataset captures changes in

rainforest environments and presents a much more challenging scenario compared to urban building CD datasets, due to subtle spectral variations, complex backgrounds, and irregular object boundaries that often result in fragmented change regions. It also exhibits significant intraclass variations caused by seasonal effects, lighting conditions, and natural vegetation dynamics. We obtain 4312 image pairs of size 256 × 256, partitioned into 2620 for training, 846 for validation, and 846 for testing. Forest disturbances are organic and irregular, which lack the clear geometric structures of built-up areas, making this dataset particularly challenging. Baseline performance on RF-CD is much lower than that seen on urban building CD datasets, such as WHU-CD and LEVIR-CD. This discrepancy highlights the inherent difficulties in detecting subtle changes in vegetation from complex and heterogeneous natural backgrounds, where weak spectral contrast and irregular spatial patterns greatly complicate the detection of such changes.

2) *Landsat-SCD* [35], which consists of 8468 bitemporal image pairs, acquired between 1990 and 2020, with spatial resolution of 30 m/pixel and 416 × 416 pixels. The dataset contains a "no-change" category and ten types of semantic change, which cover four main land-cover classes: farmland, desert, building, and water. For the forest CD task that is addressed in this study, we are specifically interested in green land and forest regions, and combine these two land cover types into one category to simplify classification objective. As no official train validation-test split is given we randomly split a subset of 1908 image pairs into training, validation, and testing sets containing 956, 476, and 476 image pairs, respectively. To be compatible with the model architecture proposed and to accommodate the computational resource constraints, all images were uniformly resized to 256 × 256 pixels before training and evaluation.
3) *SECOND* [36] is a publicly available semantic CD dataset that consists of aerial image pairs captured from a number of different cities. The dataset consists of 4662 bitemporal image pairs with a spatial size of 512 × 512 pixels, acquired from high-resolution platforms, including optical and SAR sensors. For the task of forest CD that is tackled in this study, we mainly consider green land and forest areas, i.e., tree canopy and low-lying vegetation. These two categories are combined into one class, called "forest green land," in order to match the goals of our analysis. The dataset contains pixel-level annotations manually labeled by domain experts to ensure the high-quality and reliable ground truth (GT). The dataset is randomly split into training, validation, and testing datasets with 2,671, 297, and 297 image pairs, respectively. This partitioning strategy ensures that there is enough data to train the model but still enough data to validate the model and assess performance in a unbiased manner. In order to be compatible with the proposed architecture, and also to account for computational resources, all images are resized to 256 × 256 pixels uniformly before the training and evaluation process.

Overall, we test the major performance of the proposed DECAF-CD algorithm in the forest domain using three forest-specific datasets. Furthermore, ablation studies are carried out on two commonly used public datasets that represent diverse urban scenes to check the robustness and generalizability of the key components of the framework.

4) *WHU-CD* [37] comprises high-resolution aerial imagery (0.2 m/pixel) capturing building changes in Christchurch, New Zealand, following the 2011 earthquake. The dataset presents significant challenges due to substantial variations in imaging conditions, seasonal changes, and ongoing reconstruction activities. Original images are divided into nonoverlapping 256 × 256 patches, resulting in 7487 image pairs distributed as 5947 for training, 743 for validation, and 744 for testing. This dataset is particularly valuable for evaluating model robustness against diverse environmental conditions and complex change patterns, including building demolition, construction, and modification.
5) *LEVIR-CD* [38] contains very high-resolution Google Earth imagery (0.5 m/pixel) of urban building changes across 20 regions in Texas, USA. The dataset emphasizes geometric structures and well-defined architectural changes characteristic of urban development. Preprocessing yields 10 240 nonoverlapping 256 × 256 patches, split into 7120 for training, 1024 for validation, and 2048 for testing. Its clear geometric boundaries and structured change patterns enable the evaluation of model performance on regular urban expansion and development.

B. Semisupervised Experimental Protocol

In accordance with standard semisupervised learning practices, low-annotation scenarios are simulated by randomly partitioning the training set into labeled and unlabeled subsets at four different annotation ratios: 5%, 10%, 20%, and 40% (detailed partitions in Table I). For each annotation ratio, five independent runs with different random seeds are performed to ensure statistical reliability, and report the mean performance across all runs. The validation and test sets remain fully labeled and fixed throughout for fair comparison.

C. Implementation Details

The proposed model was implemented in PyTorch using a ResNet-34 backbone pretrained on ImageNet. Input images were uniformly resized to 256 × 256 pixels. The network was optimized using AdamW optimizer with an initial learning rate of 6×10^{-4}, weight decay of 1×10^{-4}, and a polynomial learning rate decay schedule with an exponent of 0.9 was adopted. Training was conducted with a batch size of 16 comprising 8 labeled and 8 unlabeled pairs, and the network was trained for 40k iterations. Weak augmentations including random horizontal/vertical flipping and 90° rotation were applied to all samples. Following UniMatch [42], each unlabeled sample additionally received two independently sampled strong augmentations, comprising color jitter, Gaussian blur, CutMix, and random grayscale conversion. These strong augmentations were applied synchronously to both images of a given bitemporal pair to avoid

TABLE I
STATISTICS OF DATA PARTITIONS UNDER DIFFERENT ANNOTATION RATIOS IN SEMISUPERVISED SETTINGS

Dataset	Label Ratio	Labeled	Unlabeled	Val	Test
RF-CD	5%	131	2489	846	846
	10%	262	2358	846	846
	20%	524	2096	846	846
	40%	1048	1572	846	846
Landsat-CD	0.5%	10	1898	476	476
	10%	191	1717	476	476
	20%	382	1526	476	476
	40%	763	1145	476	476
SECOND-CD	0.5%	13	2658	297	297
	10%	267	2404	297	297
	20%	534	2137	297	297
	40%	1068	1603	297	297
LEVIR-CD	5%	356	6764	1024	2048
	10%	712	6408	1024	2048
	20%	1424	5696	1024	2048
	40%	2848	4272	1024	2048
WHU-CD	5%	297	5650	743	744
	10%	594	5353	743	744
	20%	1189	4758	743	744
	40%	2378	3569	743	744

introducing artificial pseudochanges. The loss weights for the CP branch were set to $\lambda_{\text{CP}_{\text{BCE}}} = 0.5$, $\lambda_{\text{CP}_{\text{Dice}}} = 0.5$, and for the consistency training (CT) branch $\lambda_{\text{CT}_{\text{FT}}} = 0.6$, $\lambda_{\text{CT}_{\text{Combo}}} = 0.4$.

The GAF module employs grouped depthwise dilated convolutions with dilation rates $\{1, 3, 5, 7\}$, while the CGC module integrates LKA using 5×5 and 7×7 depthwise convolutional kernels. A mean-teacher framework with an EMA momentum of $\alpha = 0.996$ is adopted. The consistency loss weight is linearly ramped from 0 to 2.0 over the first 8k iterations. A fixed confidence threshold of $\tau = 0.95$ is applied to teacher predictions to filter unreliable pseudolabels and suppress confirmation bias, following common practice in semisupervised learning [42]. For datasets, such as Landsat-SCD and SECOND, which contain multiple land-cover categories, the green land and forest classes are merged into a single forest change category across all methods. This ensures a consistent label space for all comparable evaluation. All baseline models were reimplemented and trained under identical data splits, augmentation pipelines, category definitions, and optimization settings to guarantee reproducibility and enable valid performance comparisons.

D. Evaluation Metrics

Five complementary metrics are employed to assess both pixel-level accuracy and instance-level quality.

Precision (P), recall (R), and $F1$-score ($F1$) serve as fundamental metrics for binary classification performance

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad F1 = \frac{2 \times P \times R}{P + R} \tag{19}$$

where TP, FP, and FN denote true positives (correctly detected changes), false positives (background misclassified as changes), and false negatives (missed changes), respectively. Precision reflects prediction reliability, recall measures detection completeness, and $F1$ provides a balanced harmonic mean that addresses precision–recall tradeoff.

Intersection over union (IoU), also known as the Jaccard index, evaluates the spatial overlap between predictions and GT

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{20}$$

IoU is a stringent metric that simultaneously penalizes both oversegmentation and under segmentation, making it a stringent measure of localization accuracy.

Panoptic quality (PQ) evaluates instance-level segmentation quality (SQ) by combining SQ and recognition quality (RQ)

$$\text{PQ} = \frac{\sum_{(p,g)\in \text{TP}} \text{IoU}(p, g)}{|\text{TP}|} \times \frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|} \tag{21}$$

where (p, g) represents matched prediction-GT instance pairs, and $\text{IoU}(p, g)$ measures their spatial overlap. SQ captures shape-alignment accuracy, while RQ evaluates detection completeness. In CD, instances are defined as connected components in the binary change maps, and compute PQ to assess the geometric integrity and structural preservation of detected change regions. This metric is particularly valuable for applications requiring precise boundary delineation, which is critical for practical applications, such as urban planning and disaster assessment.

Together, these metrics provide a multifaceted evaluation of pixel-wise accuracy, localization precision, and instance-level detection quality.

E. Comparison Algorithms

We performed comparative analysis of seven state-of-the-art CD methods, SemiCD, FixMatch, UniMatch, C2F-SemiCD, SCanNet, SEIFNet, and SARASNet, to rigorously assess the performance of the proposed approach with current advanced techniques.

1) *SemiCD* [39]: SemiCD is a semisupervised CD model that enforces consistency on the predicted change probability maps of unlabeled dual-temporal images. Small random perturbations are applied to the latent feature difference maps to encourage robustness. During the training process, both unsupervised CD loss and the supervised cross-entropy loss are computed. This consistency-based strategy effectively utilizes unlabeled data while preserving fine-grained change details.
2) *FixMatch* [40]: FixMatch is a generic semisupervised framework adapted for remote sensing CD, combining pseudolabeling and consistency regularization. The model generates pseudolabels for weakly augmented unlabeled samples and enforces consistency under strong augmentations. This principled integration of unlabeled data improves generalization in low-annotation regimes and reduces dependency on full supervision.
3) *UniMatch* [41]: UniMatch is a semisupervised CD paradigm that unifies pseudolabeling and consistency-based learning within a single framework. UniMatch dynamically generates pseudolabels for unlabeled bitemporal images, while enforcing consistency constraints on multiscale feature representations. This strategy enhances the detection of both subtle and large-scale changes by

TABLE II
COMPREHENSIVE PERFORMANCE COMPARISON ON RF-CD DATASET UNDER VARYING ANNOTATION RATIOS (%)

Method	5%					10%					20%					40%				
	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ
SupBaseline	40.2	62.5	58.8	60.6	28.5	43.8	66.2	62.1	64.1	31.2	46.5	68.9	64.8	66.8	33.5	49.2	71.0	66.1	68.5	35.8
SemiCD	44.8	67.1	64.8	65.9	32.2	47.5	70.5	67.9	69.2	34.8	51.2	73.2	69.8	71.5	37.5	53.8	75.5	70.8	73.1	39.5
FixMatch	45.5	68.2	65.5	66.8	33.5	49.2	72.8	69.2	70.9	36.8	52.9	75.1	71.6	73.3	38.8	54.9	76.6	72.6	74.5	40.2
UniMatch	44.8	68.1	63.2	65.8	32.8	49.7	69.9	68.5	69.3	36.9	53.3	70.8	71.2	70.9	39.2	56.2	71.4	72.5	71.9	40.8
C2F-SemiCD	48.5	71.1	67.0	69.0	36.8	52.7	74.5	70.8	72.6	39.5	55.6	76.2	72.4	74.2	41.2	57.5	77.5	73.5	75.4	42.5
ScanNet	47.2	68.5	68.2	68.3	34.5	50.5	69.8	69.5	69.6	37.2	53.8	71.0	71.2	71.1	39.5	56.4	71.9	72.1	72.0	41.2
SEIFNet	48.2	69.5	68.3	68.8	36.3	52.0	71.8	71.5	71.6	38.9	55.2	74.3	**73.6**	73.9	40.8	57.7	75.9	75.3	75.6	42.5
SARSNet	45.5	67.2	65.8	66.5	32.9	49.5	69.8	68.5	69.2	36.7	52.5	70.9	69.5	69.9	38.2	55.5	71.7	71.3	70.9	39.8
Ours	**49.8**	**72.8**	**68.5**	**70.6**	**37.5**	**53.3**	**75.2**	**71.8**	**73.4**	**40.1**	**56.1**	**76.8**	73.2	**75.0**	**41.5**	**58.8**	**78.9**	**75.3**	**77.1**	**43.6**

Best results in bold, second best underlined. All values represent mean over 5 runs with different random seeds.

effectively leveraging the spatial-temporal information in unlabeled data.

4) *C2F-SemiCD* [42]: C2F-SemiCD is a coarse-to-fine semisupervised CD network that progressively refines predictions using labeled and unlabeled samples. The coarse stage produces initial change maps from bitemporal inputs, a fine stage then refines them with high-resolution features and pseudolabel guidance. The framework leverages unlabeled data through consistency learning and entropy minimization, enabling robust detection of heterogeneous, and small-scale changes in complex natural scenes.
5) *SCanNet* [43]: SCanNet is a hybrid semantic CD framework that uses triple CNN and transformer. A triple encoder–decoder network is used to extract temporal features and change representations from bitemporal imagery. Following this, a semantic change transformer models spatio-temporal dependencies in the token space, and explicitly captures "from to" semantic transitions between the images. In addition, a semantic learning scheme leads the network to make use of spatio-temporal constraints and to model intrinsic correlations between changes and their semantic classes. By combining these joint spatio-temporal representations, SCanNet can significantly improve the accuracy of semantic CD tasks.
6) *SEIFNet* [44]: SEIFNet is a spatiotemporal enhancement and interlevel fusion network for CD. The framework starts by extracting multiscale feature maps from the bitemporal images using a backbone network. A spatiotemporal difference enhancement module with a dual-branch architecture then captures the global changes at the target level and fine-grained local differences between temporal features at the same scale. Enhanced features are then decoded using an adaptive context fusion module and refinement module. Finally, a summation-based prediction head combines multiscale representations to produce the final change map, which effectively fuses global and local spatio-temporal information.
7) *SARSNet* [45]: SARASNet is a scale and relation aware Siamese network for CD. First, a relation-aware module works on feature maps extracted from the two input images before feature subtraction to strengthen the interactions between the two images for better feature discrimination. Following subtraction, a scale-aware attention module calculates cross-scale attention on the difference maps, effectively handling scene changes due to objects of different sizes. Finally, a cross-transformer module is used to fuse multilevel features to highlight the spatial information, which helps to better separate the foreground and background and reduce the false positives in the change map.

F. Overall Performance Comparison

Tables II and III show a detailed comparison of the proposed method with seven state-of-the-art semisupervised CD methods on the RF-CD, Landsat-SCD, and SECOND benchmark datasets. Evaluations are performed under annotation ratios of 5%, 10%, 20%, and 40%, thus emphasizing the ability of the method in cases where different amounts of labeled data are available. All quantitative results are averaged over five independent runs with different random seeds to ensure statistical robustness and mitigate the effect of random partitioning. The three datasets together cover a variety of forest and natural vegetation settings, including primary rainforests, secondary growth forests, and mixed forest ecosystems. In particular, RF-CD is composed solely of tropical rainforest scenes, and captures natural vegetation dynamics with irregular boundaries, gradual spectral transitions, and low-contrast disturbances. From Landsat-SCD and SECOND, we derive subsets that mainly focus on changes related to forests, thus expanding the scenarios for forest CD and increasing the assessment of model performance in complex natural environments.

1) Performance on RF-CD Dataset: Fig. 5 shows the results of forest CD for 8 representative image pairs. Methods like SemiCD, FixMatch, and UniMatch completely miss most of the change regions and only capture a few isolated changes in the latter pairs. C2F-SemiCD, SCanNet, SEIFNet, and SARASNet produce more complete change maps but they still have omissions and fragmented boundaries. In contrast, the proposed DECAF-CD method can always generate results that are close to the GT, and can delineate the entire change regions with well-defined edges and little false detections in all image pairs, showing the superior ability of the proposed method to handle subtle and complex forest disturbances.

The RF-CD dataset, which is focused on detecting forest disturbance, presents great challenges because of spectral

TABLE III
COMPREHENSIVE PERFORMANCE COMPARISON ON LANDSAT DATASET UNDER VARYING ANNOTATION RATIOS (%)

Method	5%					10%					20%					40%				
	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ
SupBaseline	69.7	82.9	79.6	81.2	68.3	71.9	83.1	80.1	81.5	70.8	73.3	83.5	80.7	82.0	71.5	74.2	83.6	81.3	84.1	72.6
SemiCD	70.2	83.5	80.1	81.7	70.1	72.1	84.8	81.5	83.1	72.3	73.8	85.9	82.6	84.2	73.9	75.3	86.8	83.7	85.1	75.3
FixMatch	71.5	84.2	81.8	83.0	71.8	73.4	85.3	83.2	84.2	73.6	75.0	86.1	84.1	85.1	75.2	76.4	86.8	85.1	85.9	76.7
UniMatch	73.8	86.2	83.5	84.8	74.2	75.2	87.1	84.6	85.8	75.8	76.6	87.7	85.3	86.5	77.2	78.1	88.1	86.1	87.1	78.7
C2F-SemiCD	74.8	86.9	84.0	85.2	75.5	76.2	87.8	85.0	86.2	77.2	77.5	88.5	86.2	87.2	78.5	78.0	89.1	86.5	87.5	78.8
ScanNet	74.5	84.8	85.2	85.0	75.3	76.2	85.5	86.5	86.0	77.1	77.6	86.2	**87.6**	86.9	78.7	79.1	86.8	**88.9**	87.8	80.2
SEIFNet	75.1	86.5	84.2	85.3	75.8	76.5	87.3	85.3	86.3	77.4	77.4	87.7	85.9	86.8	78.7	78.5	87.9	86.9	87.4	79.3
SARSNet	74.2	**87.8**	82.1	84.8	74.5	75.8	**88.6**	83.5	85.9	76.2	77.0	**89.2**	84.3	86.7	77.6	78.3	**89.8**	85.0	87.2	78.6
Our	**75.6**	86.8	**85.5**	**86.1**	**76.4**	**77.1**	87.6	**86.8**	**87.2**	**78.0**	**78.3**	88.3	87.5	**87.9**	**79.3**	**79.4**	89.0	87.1	**88.0**	**80.3**

Best results in bold, second best underlined.

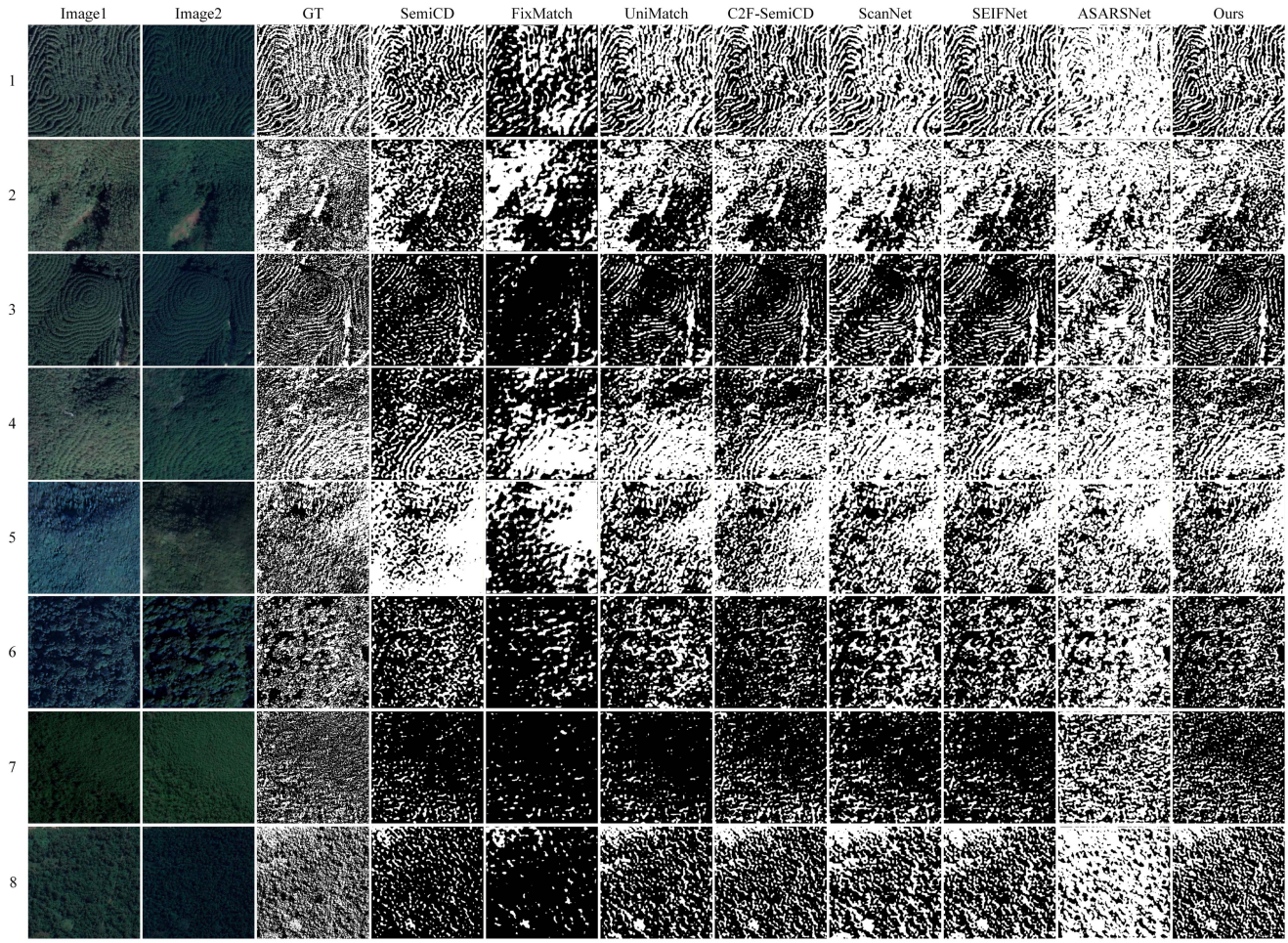


Fig. 5. Qualitative evaluation results of SOTA comparison methods with a 40% labeled ratio on the RF-CD dataset.

variability and complex texture patterns. As summarized in Table II, the proposed DECAF-CD method achieves an IoU of 49.8% under 5% annotation and 58.8% under 40% annotation, which is consistently better than all the baseline approaches. Among the compared methods, C2F-SemiCD is the best baseline and can be used as the main reference to assess the effectiveness of our model. At 5% annotation, DECAF-CD outperforms C2F-SemiCD by 1.3 pp in IoU and 1.6 points in $F1$, with an increase of 1.7 points in precision. More importantly, the method achieves the best PQ of 37.5% at 5% annotation, which is 0.7 points higher than C2F-SemiCD (36.8%) and achieves better instance-level segmentation under extremely limited supervision. This kind of comprehensive improvement is especially important in the case of forest CD, where missed disturbances may have severe ecological consequences. Performance improvements are consistent across all annotation ratios with IoU improvements ranging from 0.5 to 1.3 percentage points. At 40% annotation, DECAF-CD improves the recall by 1.8 points compared to C2F-SemiCD and also has the highest PQ of 43.6%, beating the second best method (42.5%) by 1.1 points.

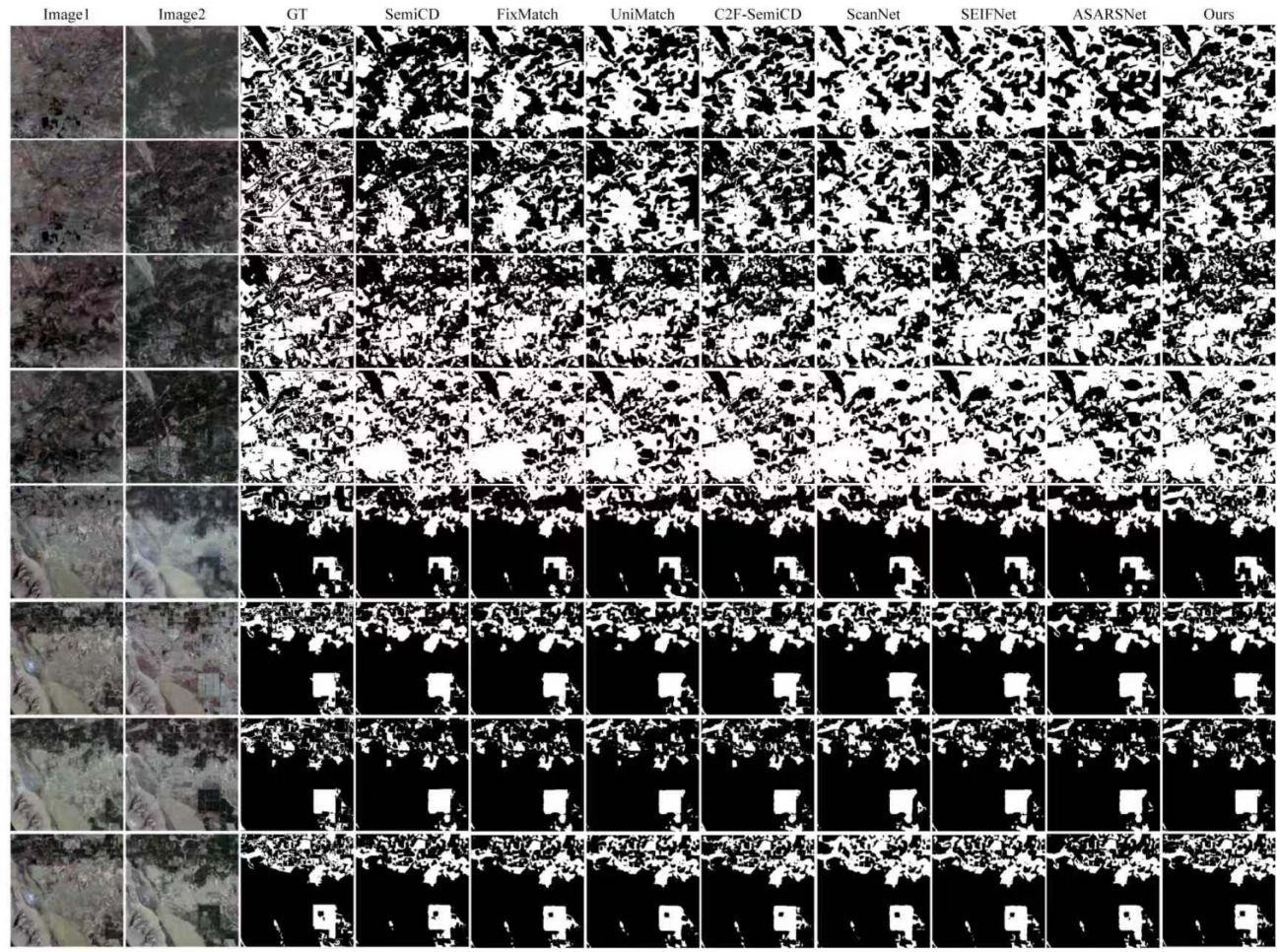


Fig. 6. Qualitative evaluation results of SOTA comparison methods with a 40% labeled ratio on the Landsat-SCD dataset.

These results highlight the robustness of the proposed framework under different levels of supervision, suggesting that the effectiveness of the framework is not limited to a particular annotation setting.

2) Performance on Landsat-SCD Dataset: Fig. 6 shows CD results on the Landsat-SCD dataset. Methods, such as SemiCD, FixMatch, and UniMatch, produce coarse boundaries and suffer from considerable detail loss. C2F-SemiCD, SCanNet, SEIFNet, and SARASNet generate more complete change maps than these methods, but still have gaps and fragmented edges. In contrast, the proposed DECAF-CD technique produces well-defined boundaries and is effective in suppressing noise, and retaining fine-grained details from all samples. It has the highest prediction accuracy consistently, especially in difficult cases, which is much better than all the compared methods and shows its robustness in capturing subtle and complex forest-related changes.

Tables III shows a detailed performance comparison on the Landsat-SCD dataset, where the proposed DECAF-CD method is consistently better than all competing approaches for any annotation ratio and evaluation metric. With only 5% annotated samples, DECAF-CD achieves IoU of 75.6% which is better than the second-best method, SEIFNet (75.1%), by 0.5 percentage points, and is clearly better than other strong baselines, such as C2F-SemiCD (74.8%) and UniMatch (73.8%). Notably, at 5% annotation, DECAF-CD achieves an $F1$ of 86.1% and a PQ of 76.4%, which is the highest among all methods, with PQ better than the second-best (75.8%) by 0.6 points. This shows excellent instance-level SQ with very limited supervision. These performance advantages continue to accrue as the ratio of annotation increases. At 40% annotation, DECAF-CD achieves 79.4% IoU, 89.0% precision, and 87.1% recall, achieving the highest $F1$ of 88.0% and the highest PQ of 80.3%. PQ exceeds ScanNet (80.2%) by 0.1 points and C2F-SemiCD (78.8%) by 1.5 points, demonstrating steady improvements in SQ despite more labeled data. The simultaneous improvement in both precision and recall in all settings suggests that DECAF-CD improves feature discriminability while preserving a tradeoff between false positive and false negative, supporting robustness in forest CD tasks.

3) Performance on SECOND Dataset: The SECOND dataset reflects changes in land-cover in a variety of environments with characteristics different from the mostly vegetation-focused Landsat-SCD dataset. Visual comparisons of the results of forest CD on SECOND are shown in Fig. 7. Methods like SemiCD, FixMatch, and UniMatch do not accurately delineate disturbance boundaries, leading to both false positive and missed detections. C2F-SemiCD, SCanNet, SEIFNet, and SARASNet

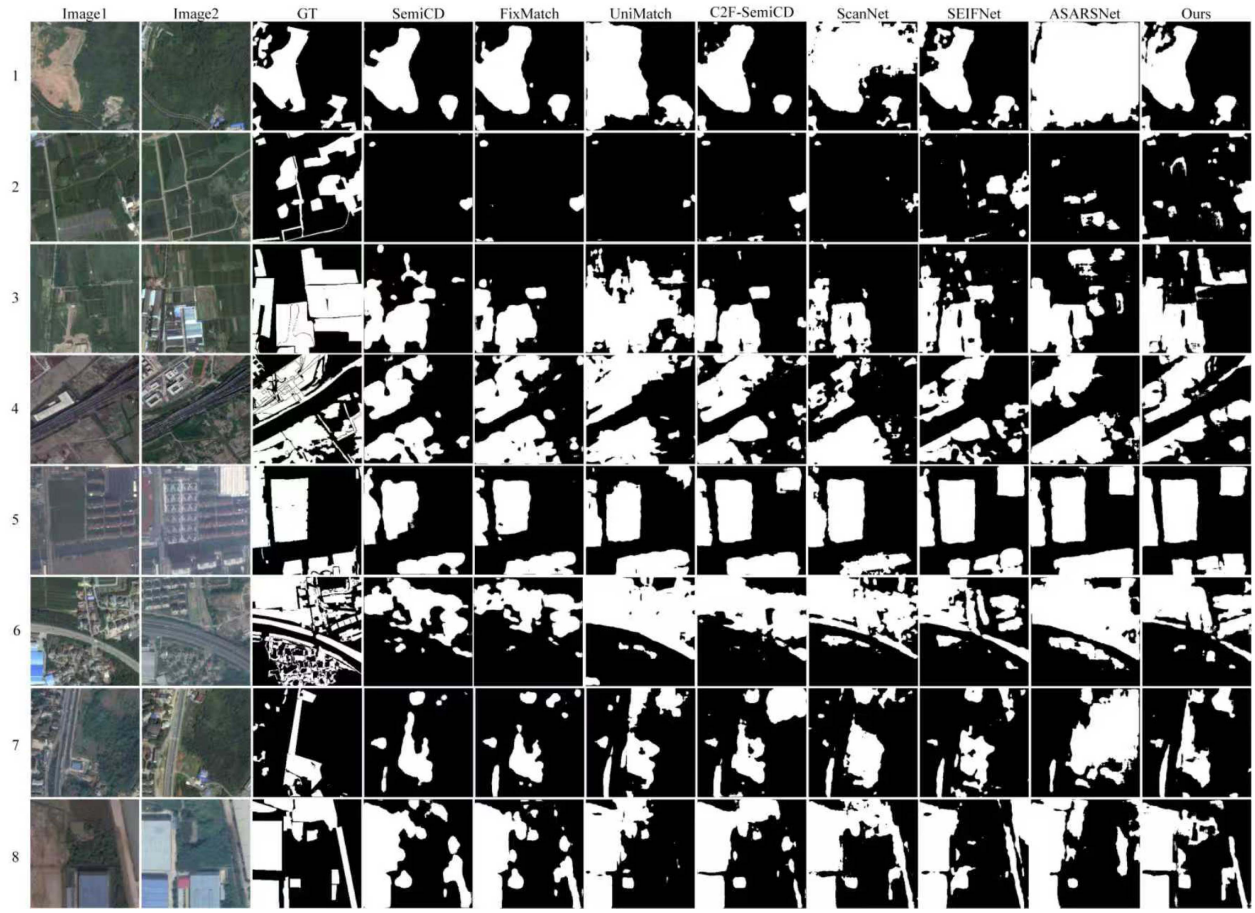


Fig. 7. Qualitative evaluation results of SOTA comparison methods with a 40% labeled ratio on the SECOND dataset.

TABLE IV
COMPREHENSIVE PERFORMANCE COMPARISON ON SECOND DATASET UNDER VARYING ANNOTATION RATIOS (%)

Method	5%					10%					20%					40%				
	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ	IoU	*P*	*R*	*F*1	PQ
SupBaseline	50.6	72.7	68.9	71.6	46.5	55.6	75.2	70.1	73.2	51.5	57.2	76.9	71.4	74.6	53.7	60.3	77.3	69.1	75.1	55.9
SemiCD	59.5	75.8	71.2	73.4	57.8	61.3	77.2	72.8	74.9	59.9	62.9	78.5	73.9	76.1	61.4	64.6	79.4	75.2	76.9	62.9
FixMatch	61.2	77.5	72.8	75.1	59.8	63.1	78.9	74.3	76.5	61.9	64.6	79.8	75.2	77.4	63.4	66.0	80.9	76.1	78.1	64.7
UniMatch	62.0	75.2	75.8	75.5	62.5	63.8	76.5	76.9	76.7	64.2	65.3	77.4	78.2	77.8	65.8	66.8	78.6	**79.4**	79.0	67.2
C2F-SemiCD	62.2	78.2	74.0	76.0	62.8	64.2	79.5	75.8	77.5	64.5	65.8	80.5	77.0	78.6	65.5	67.1	80.8	78.0	79.2	67.0
ScanNet	61.5	77.8	73.5	75.6	60.8	63.4	78.9	75.0	76.9	62.7	65.0	79.6	76.2	77.8	64.3	66.5	80.2	77.2	78.6	65.8
SEIFNet	62.8	78.9	74.2	76.5	62.2	64.7	80.1	75.6	77.8	64.1	66.3	81.0	76.8	78.8	65.8	67.6	81.8	77.6	79.4	66.9
SARSNet	58.2	77.5	69.5	73.3	55.1	60.1	78.6	70.8	74.5	57.3	61.6	79.4	71.8	75.4	58.8	63.2	80.2	72.9	75.5	60.1
Our	**63.5**	**79.5**	**75.8**	**77.6**	**63.2**	**65.4**	**80.6**	**77.1**	**78.8**	**65.1**	**67.2**	**81.5**	**78.5**	**80.0**	**67.0**	**68.7**	**81.8**	79.2	**80.3**	**68.7**

Best results in bold, second best underlined.

face challenges with complex forest structures, resulting in fragmented or misaligned change maps. In contrast, the proposed DECAF-CD method produces clean and continuous boundaries and robustly detects changes in all samples. It has the highest accuracy scores especially for challenging cases, which demonstrates its superior ability to handle heterogeneous forest patterns and complex land-cover variations.

To further quantify these improvements, we tested DECAF-CD on the SECOND dataset with different annotation ratios. As shown in Table IV, the method achieves 63.5% IoU at 5% annotation and 68.7% IoU at 40% annotation, which is significantly better than all baseline approaches. At 5% annotation level, DECAF-CD achieves a recall of 75.8%, which outperforms the second best method, SEIFNet (74.2%) by 1.6 percentage points and outperforms C2F-SemiCD (74.0%) by 1.8 points. More importantly, it reaches the best $F1$ (77.6%) and PQ (63.2%) at this annotation level, which shows a better balance between precision and recall and a better SQ on the instance level. Given the subtle and gradual nature of many changes in forests, high recall is critical to ensure that ecologically significant disturbances are not overlooked. The IoU gains over SEIFNet are stable for all annotation ratios, which are about 0.7–1.1 percentage points, while improvements over C2F-SemiCD range from 1.1 to 1.6 points, indicating the robustness of the proposed framework.

At 40% annotation, DECAF-CD achieves the highest IoU, F1, and PQ of 68.7%, 80.3%, and 68.7% respectively, compared with the other methods. The $F1$ outperforms the second-best (79.4%) by 0.9 points, which further validates the effectiveness of the method under more abundant supervision. Although the SECOND dataset has more heterogeneous and less structured change patterns compared to Landsat-SCD, DECAF-CD still provides significant performance improvements, validating its robustness across different spatial structures and complex scene types.

4) Overall Performance Discussion: Compared with other methods evaluated in our experiments, DECAF-CD brings out the benefits of a domain-aware architectural design. The proposed GAF module achieves multilevel feature integration, which is equivalent to combining low-level spatial information and high-level semantic information. This capability is critical for accurate detection of heterogeneous land-cover change in complex landsat scenes, where different types of change, including vegetation change, village relocation, and water body fluctuation, co-occur. The CGC module further increases the robustness of the model through bidirectional feature interactions, reducing the impact of pseudochanges and atmospheric perturbations that are common in multispectral satellite imagery, especially when low annotation ratios are used. Embedded within CGC, the LKA component is very effective in capturing forest disturbance patterns together with their surrounding context, due to its large receptive field. Combined with the learnable weighted fusion mechanism in CGC, DECAF-CD adaptively emphasizes features at suitable scales, which allows more accurate delineation of change boundaries and better detection of subtle, small-scale forest disturbances. This integrated design is the basis for the superior performance of the method across different datasets and annotation settings.

Collectively, these results confirm that the proposed architectural innovations substantially extend the capacity of semisupervised learning to capture spatially and spectrally complex change patterns under low-annotation scenarios, particularly in natural environments where traditional models fail to generalize.

G. Analysis of Annotation Efficiency

A major advantage of the proposed DECAF-CD method is that it achieves outstanding performance under little supervision. The performance differences between our approach and baseline methods are the largest at low annotation ratios. On RF-CD (see Tables II), DECAF-CD outperforms the best baseline, C2F-SemiCD, by 1.3 percentage points in IoU with 5% annotation, and keeps the same advantage with 40% annotation, showing consistent superiority when the amount of labeled data varies. When compared with the generic semisupervised method UniMatch, the difference is even more striking at low annotation ratios, with a 5.0-point IoU advantage at 5% annotation that narrows down to 2.6 points at 40%, underlining the effectiveness of our domain-aware design under extremely limited supervision. Similar trends are seen on Landsat-SCD (see Table III) and SECOND (see Table IV), confirming that the proposed architectural innovations are able to make good use of unlabeled samples and generalize well across diverse forest and land-cover scenarios.

The annotation efficiency of DECAF-CD is also revealed by the fact that with only 5% annotated data, it is already better than fully supervised baselines trained with 40% annotations in all three datasets. Specifically, our method achieves 49.8% IoU on RF-CD, which surpasses the 40% supervised baseline of 49.2%; 75.6% IoU on Landsat-SCD, which beats 73.8%; and 63.5% IoU on SECOND, which beats 62.9%. This translates to annotation savings of up to 87.5%, which shows the practicality of the framework for large-scale remote sensing applications where manual labeling is prohibitively expensive.

The performance saturation behavior is different in different datasets. On Landsat-SCD, DECAF-CD achieves 95.2% of its 40% annotation performance (75.6% versus 79.4% IoU) requiring only 5% annotation. On SECOND, this ratio is 92.4% (63.5% versus 68.7% IoU). On the more difficult RF-CD dataset, the ratio is 84.7% (49.8% versus 58.8% IoU), which reflects the inherent difficulty of the dataset and the continued benefit of additional annotations. Nonetheless, even on RF-CD, DECAF-CD at 5% annotation significantly outperforms all baselines at the same annotation level, which confirms its effectiveness and robustness across diverse scenarios.

H. Comparison With Different Method Categories

1) Comparison With Supervised Baseline: Compared with the fully supervised baseline, DECAF-CD achieves significant improvements under low annotation ratios for all the datasets. At 5% annotation, IoU improvements over the supervised baseline are 9.6 percentage points on RF-CD (49.8% versus 40.2%), 5.9 points on Landsat-SCD (75.6% versus 69.7%) and 12.9 points on SECOND (63.5% versus 50.6%). These consistent gains show the effectiveness of semisupervised learning for remote sensing CD in a variety of scene types.

2) Comparison With Early Semisupervised Methods: Relative to previous semisupervised methods (e.g. SemiCD), DECAF-CD is able to achieve significant improvements on all the datasets. On RF-CD at 5% annotation the IoU is improved by 5.0 points (49.8% versus 44.8%) compared with SemiCD. On Landsat-SCD, the gain is 5.4 points (75.6% versus 70.2%), and on SECOND, our method is 4.0 points better than SemiCD (63.5% versus 59.5%). These consistent improvements confirm the benefit of combining consistency regularization with domain-specific architectural modules, overcoming the shortcomings of previous approaches that rely only on pseudolabeling or adversarial approaches and find it difficult to capture the complex spectral and structural patterns in remote sensing imagery.

3) Comparison With Generic Semisupervised Methods: Although FixMatch and UniMatch achieve good performance on natural image benchmarks, their performance varies on the three remote sensing datasets. On Landsat-SCD and SECOND, the two methods have competitive results, with UniMatch achieving 73.8% and 62.0% IoU at 5% annotation, respectively. However, on RF-CD their performance decreases to 45.5% and 44.8% IoU, which is less than DECAF-CD's 49.8%. This discrepancy shows

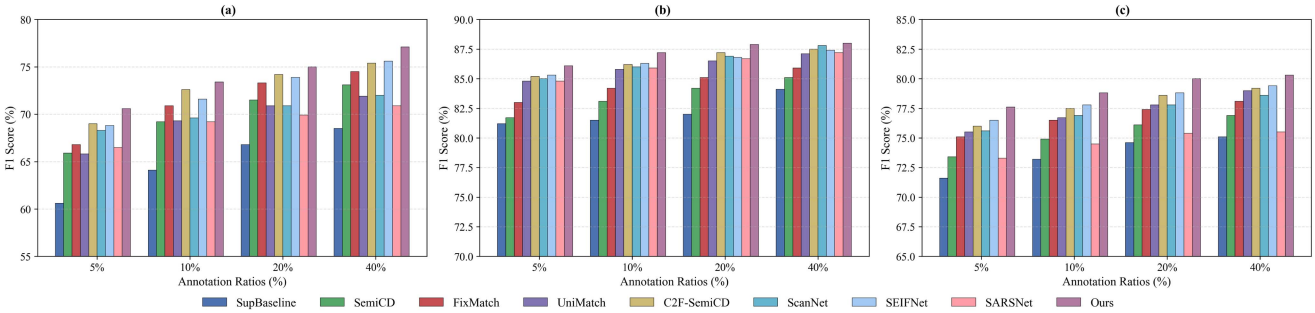


Fig. 8. $F1$ comparison of different methods on (a) RF-CD, (b) Landsat-SCD, and (c) SECOND under varying annotation ratios (5%, 10%, 20%, and 40%). Our method consistently achieves the best performance across all datasets and label regimes.

that consistency regularization strategies that have been developed for natural images classification need careful adaptation to capture subtle spectral and textural changes in complex remote sensing scenarios. The consistent improvements that DECAF-CD has achieved over these generic approaches is largely a result of the domain-aware design of DECAF-CD's modules. The GAF module is a multilevel feature integration module specifically designed for bitemporal CD, which effectively combines low-level spatial details and high-level semantic context. The CGC module with LKA offers a large effective receptive field that can capture the compact building footprint and the irregular forest patch boundary. Together, these elements make the framework adaptable to various patterns of change without requiring manual tuning for particular datasets to ensure robust performance in heterogeneous scenarios.

4) Comparison With State-of-the-Art Semisupervised Methods: Compared with the state-of-the-art semisupervised CD methods, such as C2F-SemiCD, SEIFNet, SCanNet, and SARASNet, DECAF-CD consistently outperforms all the other methods in all datasets and annotation ratios. On Landsat-SCD and SECOND, the improvements are small but consistent, with IoU improvements at 5% annotation between 0.5 and 1.4 percentage points. On RF-CD, the improvements are more pronounced with DECAF-CD outperforming these baselines by 1.3 to 4.3 percentage points in IoU at 5% annotation. The robustness of these gains across heterogeneous datasets emphasizes the generalizability of the proposed framework, showing that the effectiveness of the framework is not restricted to specific dataset characteristics or scene types.

The higher improvements seen on RF-CD can be explained by the improved capability of DECAF-CD in dealing with irregular boundaries and subtle changes. The bidirectional feature propagation in the CGC module enables information propagation between spatial locations, which is especially effective in capturing the complex spatial dependencies of natural vegetation. In addition, the learnable weighted fusion mechanism adaptively weights the features at the most relevant scales, allowing for more accurate delineation of forest boundaries than strategies with fixed fusions. This combination of adaptive and scale-aware feature integration and bidirectional propagation is the key to the superior performance of the method in challenging scenarios of forest CD.

I. Performance Visualization and Analysis

Figs. 8 –10 show in total the comprehensive performance of different methods among the three datasets and multiple annotation ratios. Fig. 8 illustrates that DECAF-CD is able to consistently achieve the highest $F1$ scores for all three datasets (RF-CD, Landsat-SCD, and SECOND) and all annotation levels, with especially strong gains at low annotation ratios. The three datasets show different levels of difficulty: Landsat-SCD has the highest $F1$ scores (81.2%–86.1% at 5% annotation), followed by SECOND (71.6%–77.6% at 5%), while RF-CD has the most difficulty (60.6%–70.6% at 5%), indicating the difficulty of identifying irregular forest boundaries in high-resolution imagery.

A unified visualization of the IoU performance across all the experimental configurations is given in Fig. 9, where darker green indicates higher values, yellow represents intermediate performance, and red corresponds to lower values. The color-coded matrix shows a clear stratification of performance, with DECAF-CD (bottom row) always having a deeper green colour for all 12 configurations. Dataset specific patterns are apparent: Landsat-SCD shows overall green-dominant colors, while RF-CD shows yellowish to reddish tones, confirming it is a more difficult dataset as a pure forest dataset. The color transition from 5% to 40% annotation is the most pronounced on RF-CD, which means this dataset is the most benefited from additional labeled data.

Fig. 10 demonstrates the superior data efficiency of our method. On RFCD, LandsatSCD, and SECOND, our method at 5% annotation (49.8%, 75.6%, and 63.5% IoU) surpasses the supervised baseline at 40% annotation (49.2%, 74.2%, and 60.3% IoU, respectively), representing annotation savings of up to 87.5% across all three datasets. The curves reveal different saturation behaviors: LandsatSCD converges rapidly after 5% annotation, SECOND exhibits gradual improvement, while RFCD maintains a steeper slope, indicating continued benefit from additional supervision on the most challenging forest scene. These visualizations collectively confirm the robustness and practical value of our semisupervised framework across diverse CD tasks.

Fig. 10 highlights the better annotation efficiency of DECAF-CD. On RF-CD, Landsat-SCD, and SECOND datesets, the method with 5% annotation (49.8%, 75.6%, and 63.5% IoU, respectively) already outperforms the fully supervised baseline

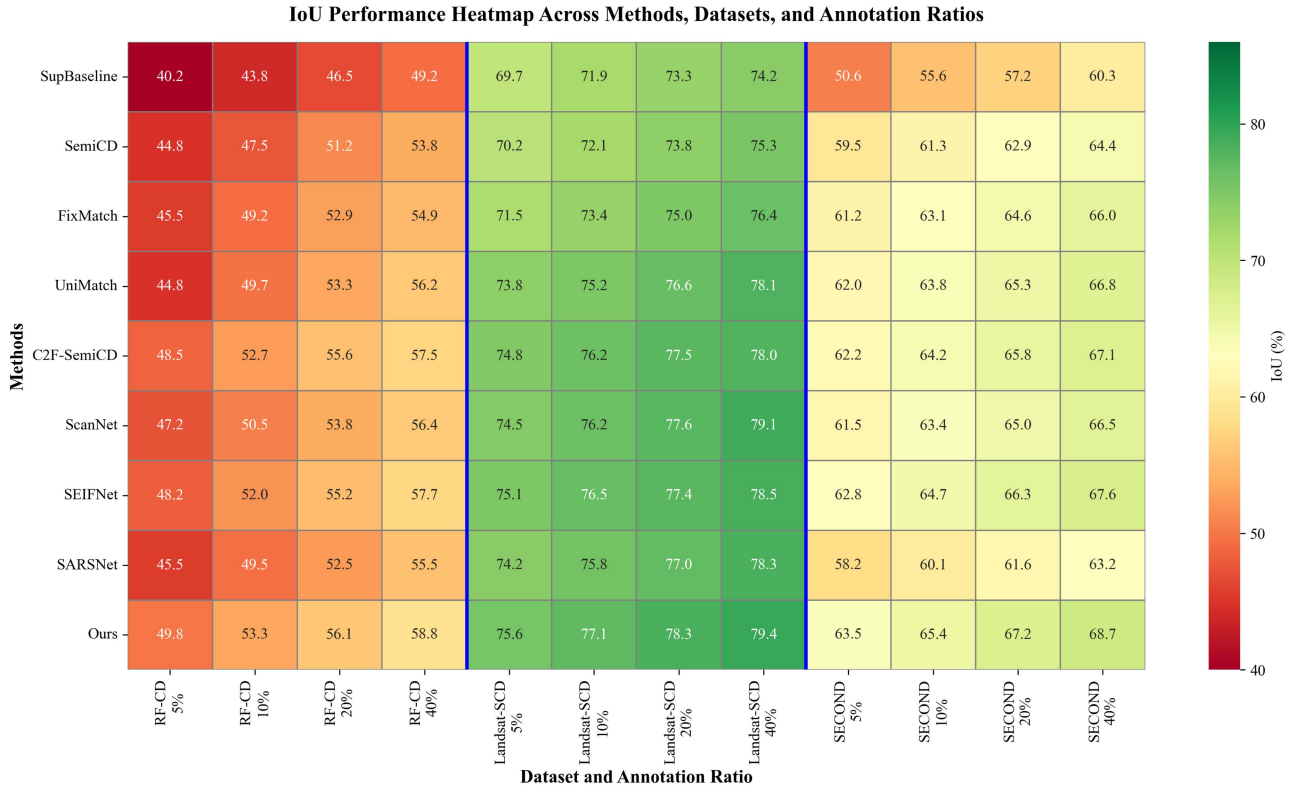


Fig. 9. Comprehensive IoU performance heatmap across nine methods, three datasets, and four annotation ratios. Darker green indicates higher IoU. Vertical blue lines separate datasets. Our method (bottom row) consistently achieves the highest IoU across all configurations.

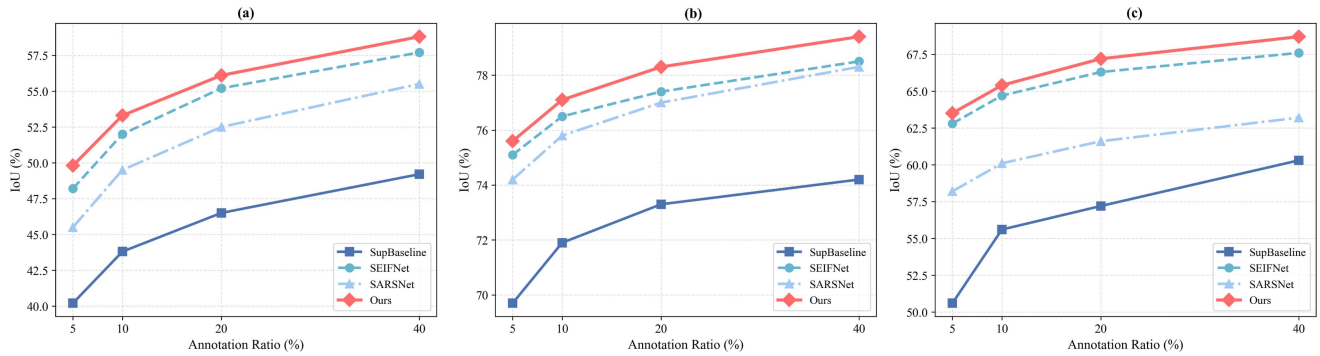


Fig. 10. Annotation efficiency analysis showing IoU versus annotation ratio for four representative methods. Our method at 5% annotation on all datasets surpasses the supervised baseline at 40%, demonstrating significant annotation savings. (a) RF-CD. (b) Landsat-SCD. (c) SECOND.

with 40% annotation (49.2%, 74.2%, and 52.3% IoU), which corresponds to annotation saving up to 87.5% across all datasets. The curves show different saturation behaviors: Landsat-SCD converges very fast after 5% annotation, SECOND shows a gradual improvement and RF-CD shows a steeper slope, indicating the persistent benefit of extra supervision for the most challenging forest scenes. Collectively, these visualizations show the robustness, data efficiency, and practical value of our semisupervised framework for a variety of CD tasks.

J. Ablation Study Analysis

To systematically assess the contribution of each of the proposed components, we performed a set of ablation experiments on the LEVIR-CD, WHU-CD, and RF-CD datasets with a 40% annotation ratio. The selection covers both structured urban environments (LEVIR-CD and WHU-CD) and the more challenging, irregular forest disturbances in RF-CD, enabling a thorough evaluation of each module's robustness and generalization. As summarized in Table V, the baseline model achieves IoU/F1/PQ scores of 81.2/90.1/72.8, 75.9/86.0/70.5, and 49.2/68.5/35.8 on LEVIR-CD, WHU-CD, and RF-CD, respectively. These results provide a stable reference point for evaluating the impact of individual modules. The significantly lower baseline performance on RF-CD (49.2% IoU) compared to LEVIR-CD (81.2% IoU) and WHU-CD (75.9% IoU) highlights the inherent challenge of forest CD, which is characterized by irregular boundaries and small spectral variations.

TABLE V
ABLATION STUDY OF THE PROPOSED COMPONENTS ON LEVIR-CD, WHU-CD, AND RF-CD (40% ANNOTATION RATIO). BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Configuration	GAF	CGC	LKA	FT+Combo	LEVIR-CD			WHU-CD			RF-CD		
					IoU	$F1$	PQ	IoU	$F1$	PQ	IoU	$F1$	PQ
Baseline	✗	✗	✗	✗	81.2	90.1	72.8	75.9	86.0	70.5	49.2	68.5	35.8
+ GAF	✓	✗	✗	✗	81.9	90.6	73.6	80.3	88.7	72.8	51.8	71.2	37.5
+ CGC (w/o LKA)	✗	✓	✗	✗	82.1	90.8	73.8	81.4	89.5	73.9	52.5	72.0	38.2
+ CGC (LKA)	✗	✓	✓	✗	82.4	91.0	74.3	82.8	90.4	75.4	54.1	73.8	39.8
+ FT+Combo	✗	✗	✗	✓	82.2	90.9	74.1	82.0	89.8	74.7	53.2	72.9	39.1
GAF + CGC (LKA)	✓	✓	✓	✗	83.0	91.2	75.4	84.6	91.5	77.4	56.5	75.4	41.8
Full (GAF + CGC + LKA + FT + Combo)	✓	✓	✓	✓	**83.8**	**91.4**	**76.5**	**85.8**	**92.7**	**79.1**	**58.8**	**77.1**	**43.6**

Each module's contribution is evaluated individually and in combination.
The bold values represent the optimal performance in the ablation study, where our proposed model DECAF-CD achieves the highest scores.

Integrating the GAF module provides consistent improvements for all the datasets with significant improvements on WHU-CD and RF-CD. On RF-CD, IoU improves from 49.2% to 51.8%, which indicates that the proposed channel-level temporal alignment mechanism improves the model's ability to capture fine-grained spatio-temporal features, thereby enabling it to better detect subtle structural changes. The relative improvement on RF-CD (+2.6 IoU points) is more significant than on LEVIR-CD (+0.7 points), which means that channel-level temporal alignment is more beneficial for complex natural scenes with heterogeneous structures.

Adding the CGC module without LKA improves performance even more on all the datasets. Notably, on WHU-CD, the $F1$ is improved from 86.0 to 89.5, and on RF-CD, IoU is improved from 49.2% to 52.5%. These improvements suggest that the bidirectional feature interaction in CGC addresses the semantic misalignment between shallow and deep layers while reducing noise caused by low-level features. Replacing the standard convolution in CGC with the LKA block yields additional gains for all the datasets. For example, PQ on WHU-CD goes from 73.9 to 75.4, and IoU on RF-CD goes from 52.5% to 54.1%, confirming that LKA is effective in modeling long-range spatial dependencies, which are important for modeling the irregular forest patch boundaries and other subtle changes in remote sensing imagery.

On the loss function side, the introduction of the Focal Tversky + Combo (FT+Combo) loss offers stable improvements on all datasets. On RF-CD, IoU is improved from 49.2% to 53.2%, which is an increase of 4.0 percentage points. This shows that Focal Tversky and Combo losses combination better addresses the class imbalance and has a better sensitivity to small or irregular change regions, which eventually improves the PQ and boundary-level accuracy. The higher improvements on RF-CD relative to more structured data sets (LEVIR-CD: +1.0 points, WHU-CD: +6.1 points) emphasize the specific usefulness of advanced loss formulations for heterogeneous natural scenes with significant class imbalance.

When GAF and CGC (with LKA) are used together, performance is significantly improved, resulting in 83.0/91.2/75.4 on LEVIR-CD, 84.6/91.5/77.4 on WHU-CD, and 56.5/75.4/41.8 on RF-CD. These results show a strong complementary effect: GAF enriches channel-level temporal alignment, and CGC with LKA provides strong multiscale contextual reasoning.

Of particular note, on RF-CD, the combined modules achieve 56.5% IoU, a 7.3 percentage point improvement over the baseline, substantially larger than that of LEVIR-CD (+1.8 points) and WHU-CD (+8.7 points). This amplified improvement demonstrates the importance of domain-aware architecture designs for challenging forest monitoring situations.

Finally, the full DECAF-CD model, which combines all the components (GAF + CGC + LKA + FT+Combo), gets the highest performance on all three datasets, which is 83.8/91.4/76.5 on LEVIR-CD, 85.8/92.7/79.1 on WHU-CD and 58.8/77.1/43.6 on RF-CD (IoU/F1/PQ). The regular improvements across diverse datasets point to the synergy between the architectural design and advanced loss formulation. On RF-CD, the full model achieves a 9.6 percentage point IoU improvement over the baseline (49.2% $\rightarrow$ 58.8%), which is significantly greater than the absolute improvement on LEVIR-CD (+2.6 points) and similar to WHU-CD (+9.9 points). Notably, the contribution of the FT+Combo loss on RF-CD (+2.3 IoU points, from 56.5% to 58.8%) is more pronounced compared to the other datasets, confirming that sophisticated loss functions are particularly effective in addressing severe class imbalance and irregular boundaries, which are key challenges in forest CD. These results collectively validate the effectiveness, robustness, and practical value of the proposed DECAF-CD framework for semisupervised remote sensing CD, in a wide range of scenarios.

K. Complexity Comparison

Fig. 11 shows a scatter plot for the computational complexity and segmentation accuracy of various methods. The x-axis is FLOPs (G), the y-axis represents IoU (%), the number of parameters (M) is represented by the point size. DECAF-CD achieves the highest accuracy (88.5% IoU) with a moderate computational footprint (150G FLOPs and 36.8 M parameters) as well as fast inference speed. SEIFNet has the lowest FLOPs (80G) with competitive accuracy (87.4%) but is still slightly behind our method. UniMatch has a reasonable balance (90G FLOPs, 87.1% IoU), while SCanNet and SARSNet have much higher computational costs (160G and 330G FLOPs, respectively) without surpassing the accuracy of DECAF-CD. Notably, the performance of the most massive model (330G FLOPs, 111 M parameters) is the lowest (85.0% IoU) in the case of the model called SARSNet. This visualization emphasizes the better efficiency-accuracy tradeoff of DECAF-CD, which shows that

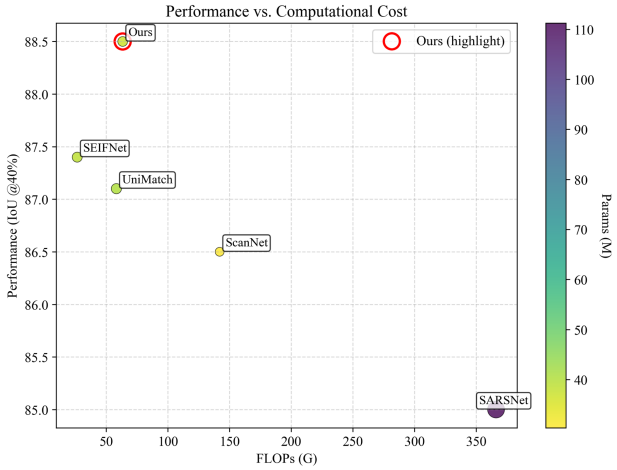


Fig. 11. Performance versus computational cost comparison. The red circle highlights our method, which achieves the highest IoU@40% with competitive parameter efficiency.

DECAF-CD achieves state-of-the-art performance with much lower computational overhead than heavier architectures.

V. Limitations and Future Work

Although the proposed framework exhibits strong robustness across datasets and annotation regimes, several limitations suggest directions for refinement. The multiscale feature aggregation and bidirectional interaction pathways introduce increased computational overhead, raising inference approximately 15% relative to the baseline. This reflects the inherent tradeoff between model expressiveness and computational efficiency, but also confirms highlights that the architectural gains are driven by modules with substantial representational capacity. Future efforts will investigate more efficient attention formulations, such as reparameterizable LKA and dynamic sparse attention, along with lightweight multibranch designs, to preserve long-range modeling ability while reducing inference latency.

A second limitation concerns the detection of extremely small change regions (often below 10 pixels), which remains challenging despite the improvements from GAF and LKA-enhanced CGC. This is especially pronounced on RF-CD, where selective logging, understory thinning, and low-intensity degradation produce subtle spatial signatures that can be lost during down-sampling operations. The difficulty in capturing these fine-grained changes implies that the current architecture, while effective at multiscale contextual reasoning, still under represents microscale structures. Future efforts could explore super-resolution-guided decoders, boundary-focused refinement heads, or adaptive high-frequency enhancement modules may further amplify the model's sensitivity to small, spatially fragmented change patterns.

A third limitation is the persistent performance gap between natural forest environments and structured urban datasets. With the full model at 40% annotation, IoU on RF-CD reaches 58.8%, substantially lower than the 83.8% and 85.8% obtained on LEVIR-CD and WHU-CD. This discrepancy does not indicate a weakness in the proposed design, but rather reflects the inherently greater complexity of forest dynamics, such as diffuse boundaries, inconsistent spectral responses, and gradual rather than abrupt changes. The analysis underscores that modules, such as GAF, CGC, and LKA, are essential for mitigating these difficulties, yet further domain-specific modeling may be required. Future work could incorporate phenology-aware feature normalization, structure-from-motion priors, or ecological-process-guided constraints to address the intrinsic irregularity of forest disturbance.

Finally, this study focuses on bitemporal CD. Although bitemporal modeling is widely adopted, real-world forest monitoring increasingly involves long-term analysis where multitemporal dependencies contain rich temporal cues about progressive degradation or recovery. Extending the semisupervised framework to recurrent or transformer-based multitemporal architectures could unlock more powerful representations of temporal trajectories. Such extensions are particularly valuable for continuous Earth observation programs, enabling more accurate modeling of disturbance propagation, regeneration cycles, and seasonal variability at scale.

Overall, these limitations do not detract from the contributions of the proposed framework; instead, they outline clear directions for advancing high-resolution forest CD.

VI. Conclusion

In this work, we proposed DECAF-CD, a semisupervised dual-encoder channel-aware network for fine-grained forest CD, designed to address the challenges of limited labeled data, heterogeneous forest structures, and subtle temporal changes in remote sensing imagery. The framework integrates a dual-encoder backbone, GAF, CGC blocks with LKA, and an MDD, DECAF-CD effectively captures multiscale, discrepancy-aware features while enhanced boundary precision and structural fidelity.

To alleviate annotation scarcity, a teacher–student semisupervised strategy leverages unlabeled bitemporal images through pseudolabel guidance and consistency regularization. Extensive experiments on RFCD, LandsatSCD, and SECOND demonstrated that DECAF-CD consistently outperforms state-of-the-art semisupervised CD methods across multiple annotation ratios, achieving higher IoU, $F1$, and PQ scores, particularly under extremely low supervision. The proposed architecture not only improves detection accuracy of complex and irregular forest disturbances but also significantly reduces the annotation effort, enabling comparable, or superior results with only 5%–10% labeled data.

Ablation studies further confirmed the contributions of GAF, CGC, LKA, and multiscale feature fusion in enhancing both global contextual understanding and local boundary refinement. Collectively, these results validate the effectiveness and generalizability of DECAF-CD for capturing fine-grained forest disturbances across diverse natural environments, offering a practical solution for large-scale, high-frequency forest monitoring where labeled data is scarce. Future work will explore the integration of multimodal data (e.g., LiDAR, SAR) to further improve detection in challenging canopy settings, as well as extend the semisupervised paradigm to continuous temporal monitoring tasks for near real-time forest management applications.

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Jinghe Zhou received the B.Sc. degree in mechanical engineering and automation from the Beijing University of Posts and Telecommunications, Beijing, China, in 2015, and the M.Sc. degree in computer science and technology from Hainan University, Haikou, China, in 2018. She is currently working toward the Ph.D. degree in computer science with the Faculty of Computing and Informatics, Universiti Malaysia Sabah, Kota Kinabalu, Malaysia.

She is currently an Associate Professor with the Hainan Vocational University of Science and Technology, Haikou, China. Her research interests include artificial intelligence (AI), ecological remote sensing, and spatiotemporal analysis.

Dr. Zhou was the recipient of Hainan's "New Star of the South China Sea" talent award, the Third Prize of the 2024 Hainan Provincial Science and Technology Progress Award, and the Second Prize of the 2024 Hainan Provincial Artificial Intelligence Science and Technology Achievement Award.



Jinmei Shi (Member, IEEE) received the Ph.D. degree in computer science from Universiti Malaysia Sabah, Kota Kinabalu, Malaysia, in 2023.

She is currently a Professor and Doctoral Supervisor with the Hainan Vocational University of Science and Technology, Haikou, China. She leads the "Computer Science and Technology" key discipline and directs the "Hainan Province Wisdom Internet of Things Engineering and Technology Research Center." She has led multiple projects funded by the National and Provincial Natural Science Foundations and has authored or coauthored more than 50 papers. Her research focuses on artificial intelligence, image analysis, and electronic information.

Dr. Shi was the recipient of Hainan's "New Star of the South China Sea" talent award, the Third Prize of the 2024 Hainan Provincial Science and Technology Progress Award, and the Second Prize of the 2024 Hainan Provincial Artificial Intelligence Science and Technology Achievement Award.



XianPeng Wang (Member, IEEE) was born in 1986. He received the M.S. and Ph.D. degrees in navigation, guidance and control from the College of Automation, Harbin Engineering University, Harbin, China, in 2012 and 2015, respectively.

He was a full-time Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from 2015 to 2016. He is currently a Professor with the School of Information and Communication Engineering, Hainan University, Haikou, China. He is the author of more than 80 papers published in related journals and international conference proceedings and was a Reviewer of more than 20 journals. His major research interests include communication system, array signal processing, radar signal processing, compressed sensing, and its applications.



Xingwang Li (Senior Member, IEEE) received the M.Sc. degree in communication engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2010 and the Ph.D. degree in communication engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2015.

From 2010 to 2012, he was with Comba Telecom Ltd., Guangzhou, China, as an Engineer. He was a Visiting Scholar with Queens University Belfast, Belfast, U.K., from 2017 to 2018. He is currently an Associate Professor with the School of Physics and Electronic Information Engineering, Henan Polytechnic University, Jiaozuo, China. His research interests span wireless communication, intelligent transport system, artificial intelligence, and Internet of Things.

Dr. Li is on the editorial board of IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE WIRELESS, COMMUNICATIONS LETTERS, IEEE COMMUNICATIONS LETTERS, etc.



Jianglin Luo received the B.E. degree in computer science and technology from the Hunan Institute of Technology, Hengyang, China, in 2022 and the M.E. degree in computer technology from Hainan Normal University, Haikou, China, in 2025.

He is currently a full-time Lecturer with the Hainan Vocational University of Science and Technology, Haikou, China. His current research interests include ecological remote sensing, and time-series analysis.



Yu-Beng Leau (Senior Member, IEEE) received the B.Sc. degree in multimedia technology from Universiti Malaysia Sabah, Kota Kinabalu, Malaysia, in 2004, the M.Sc. degree in information security from Universiti Teknologi Malaysia, Johor Bahru, Malaysia, in 2007, and the Ph.D. degree in internet infrastructures security from University Sains Malaysia, George Town, Malaysia, in 2016.

He is currently an Associate Professor with the Faculty of Computing and Informatics, Universiti Malaysia Sabah. His current research interests include intrusion detection and prediction, network security situation awareness, IPv6 security, the Internet of Things (IoT), and information centric networks (ICN).